

ConvNet Size & Parameters

Simple · STUDENT - fill in the blanks

One black-and-white image through a small ConvNet - the lecture example.

THE ONLY TWO FORMULAS (stride 1, no padding)

Conv2D: $\text{params} = ((M \times N \times L) + B) \times F$ $\text{new size} = \text{old size} - (M - 1)$
 MaxPool: $\text{params} = 0$ $\text{new size} = \text{old size} / \text{pool}$
 Flatten: $\text{params} = 0$ $\text{size} = \text{rows} \times \text{cols} \times \text{channels}$
 Dense: $\text{params} = (\text{inputs} + 1) \times \text{nodes}$ $\text{size} = \text{nodes}$

M, N = kernel size L = channels coming IN (inherited, never chosen)
 B = bias = 1 F = filters = feature maps coming OUT (you choose)

THE CODE (given)

```
model = models.Sequential()
model.add(layers.Conv2D(3, (5, 5), activation='relu',
                        input_shape=(28, 28, 1)))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(5, (3, 3), activation='relu'))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Flatten())
model.add(layers.Dense(256, activation='relu'))
model.add(layers.Dense(2, activation='softmax'))
```

Layer	Output shape	Params	Work it out - fill the blanks
Conv2D: F=3, (5x5)			size = $28 - (5-1) = \underline{\hspace{1cm}}$ params = $((5 \times 5 \times \underline{\hspace{1cm}}) + 1) \times 3 = \underline{\hspace{1cm}}$
MaxPool 2x2			size = $\underline{\hspace{1cm}} / 2 = \underline{\hspace{1cm}}$ params = 0
Conv2D: F=5, (3x3)			size = $\underline{\hspace{1cm}} - (3-1) = \underline{\hspace{1cm}}$ params = $((3 \times 3 \times \underline{\hspace{1cm}}) + 1) \times 5 = \underline{\hspace{1cm}}$ L=?
MaxPool 2x2			size = $\underline{\hspace{1cm}} / 2 = \underline{\hspace{1cm}}$ params = 0
Flatten			size = $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ params = 0
Dense(256)			params = $(\underline{\hspace{1cm}} + 1) \times 256 = \underline{\hspace{1cm}}$ size = 256 (the nodes)
Dense(2, softmax)			params = $(\underline{\hspace{1cm}} + 1) \times 2 = \underline{\hspace{1cm}}$ size = 2 (the nodes)
TOTAL trainable parameters			add up the Params column

Each row shows its formula with blanks - fill L (inherited) and the results, then check against model.summary().